

2 Experimental Methods

The liquid encapsulated electronic cooling environment was simulated inside a stainless steel pressure vessel. As shown in Fig. 1, the vessel was fitted with three ports. A module housing the simulated chip was inserted through one port, while the other two ports were fitted with quartz windows, providing optical access in directions parallel and normal to the chip surface. The fluorocarbon liquid (FC-72 or FC-87) was pre-heated by an immersion heater submerged in the liquid, and protection against heat loss to the ambient was achieved by a band heater which was wrapped around the vessel. A water-cooled condenser, coiled in the upper section of the vessel, maintained saturation conditions throughout the boiling tests. During subcooled tests, the vessel was completely filled with liquid and independent control of liquid temperature and pressure required connecting the vessel to an external pressurization/expansion tank.

Figure 2 shows a sectional diagram of the test heater module. The enhanced surfaces were built upon a 4.78 mm-thick base chip having a cross-sectional area of $12.7 \times 12.7 \text{ mm}^2$. The chip was soldered to the end of a high-purity copper bar which conducted heat from a cartridge heater to the chip. The chip heat flux was determined by a linear fit to temperatures measured by four thermocouples which were inserted along the axis of the bar. The average base temperature for the enhanced surfaces was determined by extrapolating the temperature reading of a thermocouple embedded in the chip based on the measured thermal flux. Repeatability in the data was within $\pm 0.05^\circ\text{C}$ for heat fluxes smaller than 1 W/m^2 and $\pm 0.1^\circ\text{C}$ for higher fluxes. Further details on the calibration and accuracy of the heater measurements can be found in a previous article by the authors (Anderson and Mudawar, 1989). The same article details the procedure for obtaining boiling data.

Because temperature and heat flux measurements provide little insight into the growth, size and placement of active boiling patches on an enhanced surface, visual observation played a vital role in helping to understand boiling behavior. An extensive photographic library was created to document specific boiling events for later interpretation. High speed film and digital video cameras were also used to capture brief sequences of boiling, but the most useful information came from still photography. Still photography was used in every boiling experiment to capture the changing pattern of boiling with increasing heat flux. A Nikon camera was used in conjunction with a Vivitar 55 mm automacro lens and Vivitar 152 stroblflash.

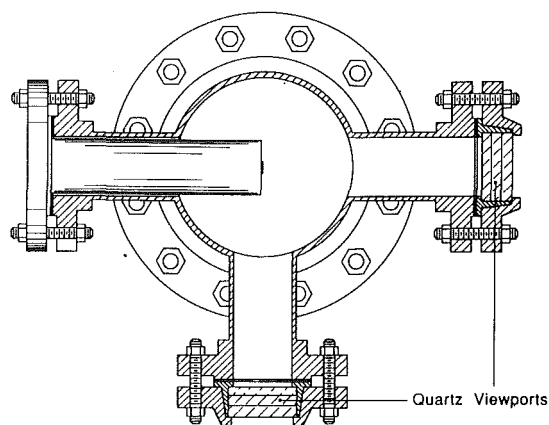


Fig. 1 Longitudinal and cross-sectional diagrams of the test chamber

Nomenclature

a = empirical constant
 A_b = base area of cylindrical fin
 A_{chip} = chip planform area ($12.7 \times 12.7 \text{ mm}^2$)
 A_e = planform area of finned element on a chip
 $A_e^+ = A_e/A_b$
 A_f = heat transfer area of cylindrical fin
 $A_f^+ = A_f/A_b$
 A_r^+ = ratio of total heat transfer area of enhanced surface to chip planform area = A_t/A_{chip}
 A_t = total heat transfer area of enhanced surface
 D = diameter of cylindrical fin
 D_b = bubble departure diameter
 g = acceleration due to gravity
 k_f = thermal conductivity of liquid

L = length of cylindrical fin
 L^+ = aspect ratio of cylindrical fin = L/D
 L_{chip} = chip length and width (12.7 mm)
 m = number of finned elements on a chip
 n = empirical constant
 Nu_D = average Nusselt number for a horizontal cylinder of diameter $D = \frac{qD}{(\Delta T)k_f}$
 Nu_l = average Nusselt number for a vertical plate of height $l = \frac{ql}{(\Delta T)k_f}$
 Pr_f = Prandtl number of liquid
 q = heat flux based on chip planform area A_{chip}
 q_b = heat flux of reference (flat,

microgroove or microstud) surface based on planform area
 Q_f = total heat transfer rate from cylindrical fin
 q_m = critical heat flux (CHF) based on chip planform area (A_{chip})
 $q_{m,\text{flat}}$ = critical heat flux from a vertical flat surface
 q_m^+ = normalized CHF value for enhanced surface
 $= \frac{q_m}{q_{m,\text{flat}}} (A_r^+)^{-1}$
 r = radial distance from axis of cylindrical fin
 Ra_D = Rayleigh number for a horizontal cylinder of diameter $D = \frac{g\beta_f(\Delta T)D^3}{\nu_f\alpha_f}$